

Rakesh Podder

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Career Objective

I am seeking a tenure-track Assistant Professor position where I can develop student-centered teaching, build independent and externally funded research program that extends my work to lead impactful, interdisciplinary research, mentor future technology scholars, and advance the state-of-the-art technology and science. I am currently a PhD candidate in the Department of Computer Science at Colorado State University (CSU), working under the mentorship of **Prof. Indrajit Ray**. My research focuses on the AI driven security & resiliency posture analysis of Cyber-Physical Systems, vulnerability management, and developing explainable and instructable AI based practical tools. Having contributed to funded projects supported by ONR, NSF, and DOE as a graduate researcher, I am positioned to pursue independent grant funding early in a faculty career. My approach to teaching is grounded in active-learning methods and building a hands-on curriculum in AI and security while mentoring the next generation of researchers and professionals in the field.

Research Interests

Artificial Intelligence • AI in Cyber Security • Large Language Models (LLMs) • Cyber-Physical System Resiliency • Vulnerability Management • Hardware Security • Formal Security Verification • Adversarial Machine Learning • Explainable AI (XAI) • Supply Chain Security • CyberEd

Education and Qualifications

2022 – present	Ph.D. Candidate Computer Science Department Colorado State University, Fort Collins, CO, USA 80521 Advisor: Professor Dr. Indrajit Ray CGPA: 4.00/4.00
2022 – 2025	Master of Science (Thesis) Computer Science Department Colorado State University, Fort Collins, CO, USA CGPA: 4.00/4.00 <i>Thesis:</i> Preventing Malicious Modifications to Firmware Using Hardware Root of Trust (HROt)
2018 – 2020	Master of Engineering (Thesis) Electrical Engineering Department Jadavpur University, Kolkata, West Bengal, India CGPA: 9.11/10 (<i>Gold Medalist</i>) <i>Thesis:</i> Data Security of IoT Enabled Autonomous Robots from the Perspective of its Various Controlled Actions
2013 – 2017	Bachelor of Technology Electrical Engineering Department Indian Institute of Technology, Patna, Bihar, India CGPA: 7.5/10 <i>Dissertation:</i> Low Cost Signal Generator

Professional Appointments

Jan. 2026 – present Colorado (USA)	Graduate Research Assistant Colorado State University Department of Computer Science
Feb. 2026 – Jun. 2026 Colorado (USA)	Graduate Mentor Office for Undergraduate Research and Artistry Colorado State University, Fort Collins, CO, USA
May 2024 – May 2026 Gaithersburg (USA)	Foreign Guest Researcher - Research and Science National Institute of Standards and Technology (NIST) Department of Computer Science
Aug 2025 – Dec. 2025 Colorado (USA)	Graduate Teaching Assistant Colorado State University Department of Computer Science
May 2022 – Aug. 2025 Colorado (USA)	Graduate Research Assistant Colorado State University Department of Computer Science
Jun. 2021 – Apr. 2022 Bhubaneswar (India)	Systems Engineer Infosys Ltd.
Nov. 2020 – May 2021 Kolkata (India)	Programmer Analyst Cognizant Technology Solutions India Private Ltd.
May 2016 – July 2016 Bokaro (India)	Summer Intern Chandrapura Thermal Power Station, DVC.

Publications

Selected Conference Proceedings

1. **R. Podder**, F. G. Arellano and I. Ray, “Hidden Amplifiers: Cross-Level Risk in Software Supply Chains,” in *Proceedings of the 42nd International Conference on Software Maintenance and Evolution (ICSME)*, 2026. arXiv:2607.05894.
2. **R. Podder** and I. Ray, “SERA – An AI Driven Re-Programmable Simulation Environment for Explainable Resiliency Analysis in Cyber-Physical Systems,” in *Proceedings of the 6th ACM Workshop on Secure and Trustworthy Cyber-Physical Systems (SaT-CPS '26)*, Frankfurt am Main, Germany, June 23–25, 2026. 10.1145/3806008.3811708
3. **R. Podder**, W. Ganim, S. Sreedharan, I. Ray, and I. Ray, “i-EXAM: Instructable and Explainable Attack Connectivity Graph Modeler,” in *Proceedings of the International Conference on Automated Planning and Scheduling (ICAPS '26)*, 2026. arXiv:2607.05888. (***Demo Paper.***)
4. **R. Podder**, J. Simental, E. Azizli, B. Mantha and I. Ray, “CoDICE: Roll the DICE for Firmware Attestation,” in *Proceedings of 2025 IEEE 7th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)*, Pittsburgh, PA, USA, 2025, pp. 183-196. 10.1109/TPS-ISA67132.2025.00028
5. T. Rhaimi, H. Aghayarzadeh, **R. Podder** and I. Ray, “Formal Specification and Verification of Protection in Transit (PIT) Protocol Using UPPAAL,” in *Proceedings of 2025 22nd Annual International Conference on Privacy, Security, and Trust (PST)*. Fredericton, NB, Canada, 2025, pp. 1-6. 10.1109/PST65910.2025.11268865
6. **R. Podder** T. Caglar, S. K. Bashir, S. Sreedharan, I. Ray, and I. Ray, “SPEAR: Security Posture Evaluation using AI Planner-Reasoning on Attack-Connectivity Hypergraphs,” in *Proceedings of the 30th ACM Symposium on Access Control Models and Technologies (SACMAT '25)*. Association for Computing Machinery, New York, NY, USA, 62–73. 10.1145/3734436.3734451
7. R. Talukdar, **R. Podder**, and I. Ray, “VKG2AG : Generating Automated Knowledge-Enriched Attack Graph (AG) from Vulnerability Knowledge Graph (VKG).” in *Proceedings of the 22nd International Conference on*

Security and Cryptography (SECRYPT 2025). SciTePress, pp. 420-428. 10.5220/0013526700003979

8. **R. Podder**, T. Rios, I. Ray, P. Raman, and S. Righi, "S-RFUP: Secure Remote Firmware Update Protocol." in *Proceedings of the International Conference on Information Systems Security (ICISS 2024)*. Lecture Notes in Computer Science, Cham, Springer Nature Switzerland, vol 15416, pp. 42-62. 10.1007/978-3-031-80020-7_3
9. S. K. Bashir*, **R. Podder***, S. Sreedharan, I. Ray and I. Ray, "Resiliency Graphs: Modelling the Interplay between Cyber Attacks and System Failures through AI Planning," in *Proceedings of the 2024 IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)*, Washington, DC, USA, 2024, pp. 292-302. (***Equal contribution.**) 10.1109/TPS-ISA62245.2024.00041 (**Best Paper Award**)
10. **R. Podder** and S. Ghosh, "Impact of White-Box Adversarial Attacks on Convolutional Neural Networks." in *Proceedings of the 2024 International Conference on Emerging Trends in Networks and Computer Communications (ETNCC)*, Windhoek, Namibia, 2024, pp. 1-9. 10.1109/ETNCC63262.2024.10767521
11. **R. Podder**, J. Sovereign, I. Ray, M. B. Santharam and S. Righi, "The PIT-Cerberus Framework: Preventing Device Tampering During Transit." in *Proceedings of the 2024 IEEE 24th International Conference on Software Quality, Reliability and Security (QRS)*, Cambridge, United Kingdom, 2024, pp. 584-595. 10.1109/QRS62785.2024.00064 (**Best Paper Award**)
12. I. Ray, S. Sreedharan, **R. Podder**, S. K. Bashir and I. Ray, "Explainable AI for Prioritizing and Deploying Defenses for Cyber-Physical System Resiliency." in *Proceedings of the 2023 5th IEEE International Conference on Trust, Privacy and Security in Intelligent Systems and Applications (TPS-ISA)*, Atlanta, GA, USA, 2023, pp. 184-192. 10.1109/TPS-ISA58951.2023.00032
13. **R. Podder** and R. K. Barai, "Hybrid Encryption Algorithm for the Data Security of ESP32 based IoT-enabled Robots." in *Proceedings of the 2021 Innovations in Energy Management and Renewable Resources(52042)*, Kolkata, India, 2021, pp. 1-5. 10.1109/IEMRE52042.2021.9386824

Selected Journals

1. **R. Podder**, M. Abdelgawad, I. Ray, I. Ray, M. Santharam, and S. Righi, "Correctness and security analysis of the protection in transit (PIT) protocol." *Journal of Systems and Software*, 230, p.112501, 2025. 10.1016/j.jss.2025.112501
2. R. Muslim, R. Delfianti, C. Harsito, S. Palaloi, N. A. Aryono, A. Y. Indrarahmana, S. R. Joshua, and **R. Podder**, "Optimizing the Integration of Wind and Solar Power for Hybrid Electrical Energy in High-Rise Building Energy Systems," *Engineered Science*, 35, p.1555, 2025. 10.30919/es1555

Selected Conference Abstracts

1. I. Ray, **R. Podder**, S. Sreedharan, I. Ray, and S. K. Bashir, "iEXAM: AI-Driven Cyber Security with Explainability." in *2026 CAE Cybersecurity Community Symposium*, Pittsburgh, Pennsylvania, 2026, pp. 73. [Abstract Link](#)
2. I. Ray, S. K. Bashir, **R. Podder**, S. Sreedharan, and I. Ray, "From Cyber Attacks to System Recovery: A Resiliency-Centered View of Cyber-Physical Systems Security." in *2026 CAE Cybersecurity Community Symposium*, Pittsburgh, Pennsylvania, 2026, pp. 74. [Abstract Link](#)

Submitted Conferences

1. **R. Podder**, V. Koscinski and I. Ray, "CAPE — Context-Aware Prioritization Engine for Vulnerability Scoring and Management," in *Proceedings of the 49th International Conference on Software Engineering (ICSE)*, 2027.
2. S. K. Bashir, P. D. Hopkins, **R. Podder**, S. Shreedharan, I. Ray and I. Ray, "AWARE-RG: Abstraction Aware Explanation of Resiliency Graphs for Cyber Physical Systems," in *the 8th IEEE International Conference on Cognitive Machine Intelligence (IEEE CogMI)*, San Jose, CA, 2026.
3. **R. Podder**, P. D. Hopkins, S. Shreedharan, I. Ray and I. Ray, "Safety Graph: When Safety Responses Become Attack Surfaces in Cyber-Physical Systems," in *Proceedings of The 8th IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (IEEE TPS)*, San Jose, CA, 2026.

Submitted Journals

1. **R. Podder**, F. G. Arellano, and I. Ray, "From Fix Commits to Vulnerable Functions: Automated Patch Mining for Function-Level CVE Resolution," in *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026.
2. S. K. Bashir, **R. Podder**, S. Sreedharan, I. Ray, and I. Ray, "AFROT: A Framework for Modeling and Analyzing Cascading Failures in CPS Triggered by Cyber Attacks," *IEEE Access*, 2026.
3. S. K. Bashir, **R. Podder**, S. Sreedharan, I. Ray, and I. Ray, "Attack Graphs, Attack Trees, and Graph Based Security Models for Cyber Physical Systems: Current Work, Open Challenges, and Future Directions," *ACM Transactions on Cyber-Physical Systems*, 2026.

Patents

1. **R. Podder**, S. K. Bashir, S. Sreedharan, I. Ray, and I. Ray, "System and Methods for Building an Instructable AI Agent for Explainable Cybersecurity Planning and Analysis," U.S. Provisional Patent Application Serial No. 63/839,787 [APPROVED]. U.S. Full Patent Application, filed on July 7, 2026 [SUBMITTED].

Awards/Honors Received

- 2026 **Computer Science Graduate Fellowship**, College of Natural Sciences, Colorado State University
- 2026 **Certificate of Academic Excellence**, Colorado State University
- 2025 Student Travel Grant, IEEE CIC/TPS/CogMI Conference
- 2025 **Robert B. France Fellowship** in Computer Science, Department of Computer Science, Colorado State University
- 2025 GSC Conference Travel Grant, Graduate Student Council, Colorado State University
- 2025 **Excellence in Teaching** Award, Colorado State University
- 2024 **Best Paper Award**, IEEE TPS-ISA Conference — [Resiliency Graphs: Modelling the Interplay between Cyber Attacks and System Failures through AI Planning](#)
- 2024 **Best Paper Award**, IEEE QRS Conference — [The PIT-Cerberus Framework: Preventing Device Tampering During Transit](#)
- 2024 GSC Conference Travel Grant, Graduate Student Council, Colorado State University
- 2024 GSC Conference Travel Grant, Graduate Student Council, Colorado State University
- 2023 Certificate of Academic Excellence, Colorado State University
- 2018 AICTE Post Graduate Scholarship, All India Council for Technical Education, Indian Government
- 2014 Top Class Undergraduate Scholarship, Ministry of Human Resource Development (MHRD), Indian Government
- 2013 Dr. B R Ambedkar Award, West Bengal State Government, India

Academic Experience

Research Experience

Colorado State University

Graduate Research Assistant

Fort Collins, CO, USA

May 2022 – Present

• CPS security and resiliency analysis using AI planning:

- *i-EXAM* – attack-connectivity graph modeling and hardening analysis for IT Network; developed [i-EXAM](#), an AI-planning and LLM-powered tool enabling system administrators to build network security profiles and run what-if hardening analyses.
- *Resiliency Graph (RG) / Safety Graph (SG)* – unified frameworks modeling how cyber attacks propagate into system failures across IT & OT domains and resiliency in cyber-physical systems.

- *SERA* – an AI-driven, re-programmable simulation environment for explainable resiliency analysis, supporting both offensive/defensive security research and a teaching module for cybersecurity education.
- **CAPE (Context-Aware CVE Prioritization Engine)** – a vulnerability analysis framework that enriches each vulnerability scanner report of vulnerabilities (CVEs) with deployment-specific evidence, then creates an AHP-weighted priority score.
- **OSCAR (Open Supply-Chain Assurance & Resilience)** – a cross-level risk propagation framework linking code-level risk metrics to ecosystem-level dependency exposure via a unified risk formula, to surface false-positive CVE alerts and structurally critical micro-dependencies otherwise missed by Software Composition Analysis (SCA) tools.

American Megatrends Inc.

Graduate Researcher

Fort Collins, CO, USA

May 2022 – Aug 2025

- **Firmware security infrastructure** – Worked with [American Megatrends Inc.](#) to design and open-source a secure firmware remote update framework (S-RFUP), a protection-in-transit protocol (PIT), and a CDDL mapping schema for firmware manifests (CoDICE) built on Microsoft Project Cerberus and Google Open-DICE.

National Institute of Standards and Technology (NIST)

Foreign Guest Researcher - Research and Science

Gaithersburg, MD, USA

May 2024 – May 2026

- Worked with [Irena Bojanova](#) on the [Bugs Framework \(BF\)](#), applying its formal language to causally classify security bugs and formalize CVE/CWE specifications.

Teaching Experience

Teaching and lab-design experience in computer networks and cybersecurity at the undergraduate and graduate level, alongside graduate coursework spanning AI, computer architecture, and cyber-physical systems.

Colorado State University

Graduate Teaching Assistant

Fort Collins, CO, USA

Aug 2025 – Dec 2025

- Teaching assistant for CS 457 (Computer Networks and the Internet); assisted with course design and led recitation sessions.
- Delivered course lectures independently, with faculty authorization, in the instructor's absence on 4 occasions.
- Designed exam questions and grading for CS 457, contributing directly to course assessment.
- Led a weekly Guidance, Study and Tutorial (GST) session during office hours: revised lecture material and worked through problem sets with students.
- Designed labs for CS 557 (Advanced Networking), including penetration-testing exercises, binary reverse-engineering labs, and other hands-on offensive-security exercises.
- At CS@CSU, I have developed an experimental research platform called *SERA* which we plan to adapt for cybersecurity exercises in undergraduate and graduate courses, in CS 356 (System Security), CS 456 (Modern CyberSecurity), CS 556 (Computer Security) and CS 557.
- Supported student research and project development in CS556 (Computer Security).
- Guided graduated student as external observer on "How to handle peer-review process and analyses research paper?" for [Usable Formal Methods for Security/Privacy](#), CS 580B4.
- Mentored more than 10+ undergraduate and graduate students on research methods, presentation skills, and publication preparation.
- Awarded *Excellence in Teaching* by CSU for applying active-learning techniques grounded in Bloom's Taxonomy to recitation and lab design.

Office for Undergraduate Research and Artistry (OURA), CSU

Graduate Mentor

Fort Collins, CO, USA

Feb 2026 – Jun 2026

- Guided undergraduate mentees through research projects, provided academic and research advising, and supported development of critical-thinking and research skills.

Cybersecurity Camp Summer Program

Fort Collins, CO, USA

Instructor / Camp Designer

2023 – 2025

- Designed lectures, hands-on activities, and classroom exercises for week-long cybersecurity day camps serving 35–50+ high school students annually across three years.
- Gained direct experience managing classroom-scale instruction and adapting technical material for varied skill levels.

Pedagogical training: Completed CSU Institute for Learning and Teaching certifications in Best Practices in Teaching, Critical Thinking Development, Inclusive Pedagogy, and Accessible Electronic Content – applied directly to course, lab, and mentoring design above (see Scientific Trainings for full list).

Industrial Experience

AMI US Holdings Inc.

Fort Collins, Colorado

Summer Intern

May 2023 – Aug 2023

Directed research to design 100+ firmware update APIs (lib. “S-RFUP”) with privileged access controls and rollback protection, 20+ (lib. “PIT”) secure boot APIs enforcing identity-based device authentication and “Co-DICE” framework for CDDL mappings of firmware manifest in BMC.

Infosys Ltd.

Bhubaneswar, India

Systems Engineer

Jun 2021 – Apr 2022

Transformed data between formats (30+ mapping tree) using IBM WebSphere, led team to optimize data handling efficiencies. Delivered more than 100 client requests of customised maps for various data formats. Taught and provided lectures for Junior Systems Engineers (Knowledge Transfer Program) about data handling and modeling.

Cognizant Technology Solutions India Private Ltd.

Kolkata, India

Programmer Analyst

Nov 2020 – May 2021

Specialized in Selenium for automated test planning and quality assurance in software development across 10+ projects.

Mentoring Experience

Rafael Fabian Gonzalez Arellano	Grad Student Computer Information Systems (M.C.I.S.) CSU (*2 Research Paper)
Wadia Ganim	Engineer Illinois ARCS Singapore (*1 Research Paper)
Sachin Kurup	CSU Intern BTech Computer Science Amrita Vishwa Vidyapeetham
Matthew Leman	Undergraduate Computer Science Department CSU
Saswat Subhankar	Undergraduate Intern Computer Science Department CSU
Enock Apedjinou	OURA Mentee Computer Science Department CSU
Jason Simental	Undergraduate Researcher Computer Science Department CSU (*1 Research Paper)
Tyler Rios	Graduate Student Computer Science Department Denver University (*1 Research Paper)
Jack Sovereign	Software Engineer Ductus (*1 Research Paper)

Presentations

Conference Presentations

1. *i-EXAM: Instructable and Explainable Attack Connectivity Graph Modeler*. **Location:** The 36th International Conference on Automated Planning and Scheduling (ICAPS 2026), Dublin, Ireland, 2026.
2. *CoDICE: Roll the DICE for Firmware Attestation*. **Location:** The 2025 IEEE 7th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA), Pittsburgh, PA, USA, 2025.
3. *Resiliency Graphs: Modelling the Interplay between Cyber Attacks and System Failures through AI Planning*. **Location:** The 2024 IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA), The Darcy, Washington, DC, USA, 2024.
4. *Impact of White-Box Adversarial Attacks on Convolutional Neural Networks*. **Location:** The 2024 International Conference on Emerging Trends in Networks and Computer Communications (ETNCC), Windhoek, Namibia, 2024.
5. *The PIT-Cerberus Framework: Preventing Device Tampering During Transit*. **Location:** The 2024 IEEE 24th International Conference on Software Quality, Reliability and Security (QRS), The Churchill College, Cambridge, United Kingdom, 2024.
6. *Hybrid Encryption Algorithm for the Data Security of ESP32 based IoT-enabled Robots*. **Location:** The 2021 Innovations in Energy Management and Renewable Resources, Kolkata, India, 2021.

Poster Presentations

1. *Correctness and Security Analysis of the Protection In Transit (PIT) Protocol*. **Location:** NSF IUCRC Center for AI-ML Driven Research in Infrastructure Trust Assurance (AMRITA) Meeting, University of Wyoming (UW), Laramie, WY, 2025.
2. *Preventing Malicious Modifications to Firmware using Hardware Root of Trust (HROt)*. **Location:** NSF IUCRC Center for AI-ML Driven Research in Infrastructure Trust Assurance (AMRITA) Meeting, University of Wyoming (UW), Laramie, WY, 2025.
3. *S-RFUP: Secure Remote Firmware Update Protocol*. **Location:** Graduate Student Showcase, Graduate School, Colorado State University, Fort Collins, CO, 2025.
4. *Explainable AI for Cybersecurity and Resiliency of Nuclear Power Plant*. **Location:** The 4th Big Data for Nuclear Power Plants Workshop, Columbus, Ohio, 2024.
5. *Correctness and Security Analysis of Protection in Transit (PIT) Protocol*. **Location:** Graduate Student Showcase, Graduate School, Colorado State University, Fort Collins, CO, 2024.
6. *How "PIT" Libraries Work?*. **Location:** NSF IUCRC Center for Cybersecurity Analytics and Automation (CCAA) IAB Meeting 2024, George Mason University, Fairfax, VA, 2024.
7. *Preventing Device Tampering During Transit using the Cerberus PIT Library and BIOS/BMC Boot Process Lockdown*. **Location:** NSF IUCRC Center for Cybersecurity Analytics and Automation (CCAA) IAB Meeting 2023, George Mason University, Fairfax, VA, 2023.
8. *Preventing Firmware Tampering during Transit*. **Location:** NSF IUCRC Center for Cybersecurity Analytics and Automation (CCAA) IAB Meeting 2023, Colorado State University, Fort Collins, CO, 2023.
9. *What is Project Cerberus?*. **Location:** Graduate Student Showcase, Graduate School, Colorado State University, Fort Collins, CO, 2023.
10. *Security Concerns for TPM 2.0*. **Location:** CS793 Research Seminar Presentation, Computer Science Department, Colorado State University, Fort Collins, CO, 2023.
11. *Implementing Protection in Transit*. **Location:** NSF IUCRC Center for Cybersecurity Analytics and Automation (CCAA) IAB Meeting, University of North Carolina at Charlotte (UNCC), Charlotte, NC, 2022.

Invited Talks

July 2026	CAE Community of Practice Cyber Research. <i>i-EXAM. A running case study on Enterprise IT Network.</i>
March 2026	Universitas Sam Ratulangi, Manado, Indonesia. <i>From Classical Systems to Intelligent Secure and Data-Centric Computing.</i>
October 2025	Northern Colorado ISSA, Lory Centre CSU, Fort Collins, CO, USA. <i>AI-Powered Cyber Security & Cyber Resiliency.</i>
July 2024	TMI Summer Talks, Fort Collins, CO, USA. <i>The PIT-Cerberus Framework: Preventing Device Tampering During Transit.</i>
October 2023	NSF IUCRC Center for CCAA Colorado State University, Fort Collins, CO, USA <i>Remote Firmware Updates: Preventing Firmware Tampering.</i>
April 2023	NSF IUCRC Center for CCAA George Mason University, Fairfax, VA, USA <i>Protection in Transit – What do we solve?.</i>
October 2022	NSF IUCRC Center for CCAA University of North Carolina at Charlotte (UNCC), Charlotte, USA <i>Protection in Transit: Enabling Tripartite Trust using a Mesh of Trusted Agents.</i>

Reviewer Services

Ad-hoc Reviewer – Journals

1. ARRAY, 2026 [**1 Paper**].
2. Journal of Information Security and Applications, 2026 [**1 Paper**].
3. Pragmatic Cybersecurity, Online ISSN: 3083-2454, 2026 [**1 Paper**].
4. IEEE Transactions on Big Data (IEEE TBD), 2026 [**1 Paper**].
5. Journal of Electronic Testing Theory and Applications (JETTA), 2025 [**1 Paper**].
6. Journal of Nonlinear, Complex and Data Science (JNCDS), 2024 [**1 Paper**].

Ad-hoc Reviewer – Conferences

1. The 49th International Conference on Software Engineering (ICSE27) Dublin, Ireland, April 25-May 01, 2027 [**3 Papers**].
2. The 3rd International Conference on Electrical and Computer Engineering Researches (ICECER'26), Istanbul, Türkiye, December 03-05, 2026 [**1 Paper**].
3. The 6th International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME'26) , Bali, Indonesia, October 15-17, 2026 [**1 Paper**].
4. The 31th European Symposium on Research in Computer Security (ESORICS'26), Rome, Italy, September 14-16, 2026. [**4 Papers**].
5. The 31st ACM Symposium on Access Control Models and Technologies (ACM SACMAT'26), Waterloo, Canada, July 8 - 10, 2026 [**3 Papers**].
6. The 6th International Conference on Electrical, Computer and Energy Technologies (ICECET'26), Rome, Italy, July 6-9, 2026 [**2 Papers**].
7. The 16th ACM Conference on Data and Application Security and Privacy (CODASPY'26), Frankfurt am Main, Germany June 23 – 25, 2026 [**1 Paper**].
8. The 3rd International Conference on Artificial Intelligence, Computer, Data Sciences and Applications

- (ACDSA'26), Boracay Island, Philippines, February 05-07, 2026 [[2 Papers](#)].
9. Workshop on Advances in Systems and Computation at the Edge and HPC Networks (ASCENT'25), ICDCN'26, Nara, Japan, January 06-09, 2026 [[2 Papers](#)].
 10. The 2025 IEEE 7th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA), Pittsburgh, USA, November 11-14, 2025 [[1 Paper](#)].
 11. The 2nd International Conference on Electrical and Computer Engineering Researches (ICECER'25), Antananarivo 101, Madagascar, December 06-08, 2025 [[2 Papers](#)].
 12. The 2025 International Conference on Emerging Trends in Networks and Computer Communications (ETNCC'25), Windhoek, Namibia, August 05-07, 2025 [[1 Paper](#)].
 13. The International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA'25), Antalya, Türkiye, August 7-9, 2025 [[2 Papers](#)].
 14. The 5th International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME'25), Zanzibar, Tanzania, October 16-19, 2025 [[1 Paper](#)].
 15. The 30th European Symposium on Research in Computer Security (ESORICS'25), Toulouse, France, September 22-26, 2025. [[3 Papers](#)].
 16. The 29th European Symposium on Research in Computer Security (ESORICS'24), Bydgoszcz, Poland, September 16-20, 2024. [[1 Paper](#)].
 17. The 14th ACM Conference on Data and Application Security and Privacy (CODASPY'24), Porto, Portugal June 19–21, 2024. [[1 Paper](#)].
 18. The 28th European Symposium on Research in Computer Security (ESORICS'23), The Hague, The Netherlands, September 25-29, 2023. [[1 Paper](#)].
 19. The 5th International Conference on Information Science and Systems (ICISS'22), Beijing, China, August 26-28, 2022 [[1 Paper](#)].
 20. The 36th Annual IFIP WG 11.3 Conference on Data and Applications Security and Privacy (DBSec'22), Rutgers University, Newark, NJ, USA, July 18-20, 2022 [[1 Paper](#)].

Public Engagement

Apr. 2027 (upcoming)	Serving as Shadow PC in 49th International Conference on Software Engineering (ICSE 2027) , Apr. 25- May 1, 2027 in Dublin, Ireland.
Jan. 2027 (upcoming)	Serving as a Technical Program Committee (TPC) Member for the 2nd ASCENT Workshop (Advances in Systems and Computation at the Edge and HPC Networks), January 04-07, which was held in conjunction with the ICDCN Conference, January 04-07, 2027.
Mar. 2027 (upcoming)	Serving as a Technical Program Committee (TPC) Member for the PerRobotics 2027: Pervasive Robotics – Sensing, Intelligence, and Coordination for Human-Centered Smart Environments , a workshop co-located with IEEE PerCom 2027 in Goa, India.
Mar. 2027 (upcoming)	Serving as Registration Chair in 30th Annual Software Technology Conference (STC 2027) , May 4-6, 2027.
Jul. 2026	Participated in Human-AI ICS Cybersecurity Challenge Research Study from July-August 2026, conducted by Nemesys Insights, LLC .
Jul. 2026	Served as Artifact Evaluation Program Committee Member in The 31st ACM Symposium on Access Control Models and Technologies (ACM SACMAT 2026), Waterloo, Canada, July 8 - 10, 2026.

Jan. 2026	Served as a Technical Program Committee (TPC) Member for the ASCENT Workshop (Advances in Systems and Computation at the Edge and HPC Networks) , January 06-09, which was held in conjunction with the ICDCN Conference, January 06-09, 2026.
Nov. 2025	Served as Student Volunteer in IEEE TPS 2025 , Nov. 2025.
Mar. 2025	Served as Review Committee Member in American Journal of Computer Science and Technology , Mar. 2025.
Jul. 2025	2025 Cybersecurity Camp Summer Program , July 2025 Hosted a one-week (5-day) CyberSecurity Day Camp at CSU Fort Collins and another one-week (5-day) Camp at CSU SPUR Campus for 50+ high school students.
Jul. 2024	2024 Cybersecurity Camp Summer Program , July 2024 Organized a one-week (5-day) CyberSecurity Day Camp at Colorado State University, Fort Collins, Colorado for 45+ high school students.
Jul. 2023	2023 Cybersecurity Camp Summer Program , July 2023 Organized a one-week (5-day) CyberSecurity Day Camp at Colorado State University, Fort Collins, Colorado for 35+ high school students.

Professional Affiliations

NCAE-C PhD Student Club

- CAE Community of Practice Cyber Research, PhD student member, since May 2026

ACM – Association for Computing Machinery

- Membership ID 8520334, since July 2025

IEEE – The Institute of Electrical and Electronics Engineers

- IEEE Member, Membership ID: 100267751, since May 2024
- IEEE Young Professionals (Affiliation)
- IEEE Systems Council (Affiliation)
- IEEE Computer Society Member (Affiliation)

Northern Colorado Information Systems Security Association (ISSA)

- NoCo ISSA Member

Institute for Systems and Technologies of Information, Control, and Communication (INSTICC)

- *Affiliate Member*

Scientific Trainings

May 2026 – July 2026 (Online)	Best Practices in Teaching at CSU TILT, Colorado State University • 4-week Course: Best Practices in Teaching • 4-week Course: Critical Thinking • 4-week Course: Inclusive Pedagogy • 4-week Course: Accessible and Inclusive Electronic Content • 4-week Course: Inclusive Pedagogy 2
May 2026 (Online)	Graduate Student Mentor Training The Institute for Learning and Teaching Colorado State University

Feb. 2026 – July 2026
(Online)

Early Career Grants Program

CSU WRITES, Colorado State University

- Webinar: Grant Program Information Session
- Webinar: RAO Grant Writing Overview – Early Career
- Webinar: Grants & GenAI - Strategies and Guidelines
- Webinar: Structuring Your Project Narrative with RAO
- Webinar: NSF Career Proposal Strategies

Jan. 2026
(Online)

Graduate Teaching Assistant Training

The Institute for Learning and Teaching
Colorado State University

Oct. 2025
(In Person)

Graduate Student Mentoring Training

The Graduate School
Colorado State University

Oct. 2025
(Online)

Excellence in Teaching

The Language, Culture, and Resources program for Excellence in Teaching
Colorado State University

Jul. 2025
(Online)

Social / Humanistic / Behavioral Research

CITI Program
Colorado State University

Aug. 2023 – Nov. 2023
(In Person)

Ethical Conduct of Research

The Graduate School
Colorado State University

Skills

Generative AI & LLMs	Claude (Sonet, Opus, Fable, Haiku), Claude Code & Coworker, ChatGPT, Ollama, Kimi, Gemini, Grok
Programming & Development	Proficient: C/C++, PDDL, Python. Familiar: Go, JavaScript, SQL, Java, Git, Verilog HDL, $\LaTeX$
Libraries and ML Algorithms	TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, Scikit-Learn, Seaborn, ART, cleverhans; LLM, CtmNN, Decision Trees, Transformers
Databases & Operating Systems	PostgreSQL, MySQL, MongoDB; Kali, Linux, Mac OS, Windows, Hypervisors, QEMU, Virtual Box, VMware, Docker
Tools & Technologies	Nmap, Nikto, Metasploit, Splunk, ELK, CAN, STRIDE, GNS3, TLA+, Wazuh, Caldera, OPNsense, pfSense, CPN, Wireshark, Google Colab, Heroku, Azure, Jupyter Notebook, UPPAAL, Syft, x86, ARM, MITRE ATT&CK, Suricata, Procmon, Grype, OpenVAS, Tenable, Active Directory, LLVM, NIST 800-61, ISO 27035, REMnux, Burp Suite
Open Source Projects	OpenCode, OpenUI, Openclaw, HexStrike AI, Project Cerberus, OpenBMC-libpldm, VAL, Forbid Iterator, Fast Downward, Open-DICE

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